Interlaboratory Evaluation of a Material with Unequal Numbers of Replicates

John Mandel and Robert C. Paule
National Bureau of Standards, Washington, D. C. 20234

Frequently, a "best value" has to be estimated on the basis of measurements from different sources, such as different laboratories. When the number of replicate measurements varies from source to source, a problem of proper weighting arises. The usual formulas cover only one or the other of two extreme situations: equal weighting of individual measurements, or equal weighting of source averages. The problem is described in detail, including the interpretation of the analysis of variance, and an iterative procedure is presented for obtaining a properly weighted average. Formulas are also given for the estimation of the uncertainty of the weighted average. An illustrative example is presented.

A FREQUENT PROBLEM in data analysis is to estimate a "best value" from a set of measurements subject to two or more sources of variability. To cite a concrete example, suppose that each of k different laboratories analyze the same material, and that we wish to derive an "optimum" estimate of the true value from the results obtained by all laboratories. It is not always possible, in such cases, to follow the theoretically very desirable procedure of having each participating laboratory make the same number of replicate determinations. The resulting lack of balance introduces some difficulties, both in the estimation of the two components of variability—within and between laboratory variance—and in the determination of the "best" overall average.

The problem is particularly important in round robin testing, when a "best value" must be derived on the basis of results obtained in a number of participating laboratories.

In this paper we discuss these difficulties and propose a procedure for dealing effectively with this problem.

The Model. Let y_{ij} represent the jth result obtained in the ith laboratory. If the number of laboratories is k, we have

$$i = 1, 2, \ldots, k$$

We suppose that the number of measurements contributed by the ith laboratory is n_i. Thus, we have

$$j = 1, \ldots, n_i \qquad \text{for laboratory } i.$$

Our assumed model is

$$y_{ij} = \mu + L_i + \epsilon_{ij} \tag{1}$$

where μ is a constant representing the "true value" (assuming that the measuring process is without bias); L_i is the "laboratory effect" (or laboratory bias) of the ith laboratory; and ϵ_{ij} is the "replication error" (or "within laboratory" error) due to the jth measurement of laboratory i.

The values of L_i are assumed to represent a random sample from a population of mean zero and standard deviation σ_B (B for between laboratories). The values of ϵ_{ij} (all i and all j) also are assumed to be a random sample from a population of zero mean. The standard deviation of this population is denoted by σ_W (W for within laboratories). The two variances σ_W^2 and σ_B^2 are the components of variance due, respectively, to within laboratory and between laboratory variability. To the extent that the F distribution is used in the subsequent analysis, the usual assumptions of approximate normality are made.

Analysis of Variance. It is customary to first perform an analysis of variance as shown in Table I, [(*1*) appendix 5C]. The last column, entitled "E(MS)," gives the expected values of the mean squares, under the assumption that the "within laboratory" variability, *i.e.*, the variability between replicate measurements made in the same laboratory is, theoretically, the same for all laboratories. This is not an unreasonable assumption when all laboratories follow the same procedure and are all competent in carrying it out. Otherwise, the assumption must be checked, *e.g.*, by making a test of "homogeneity of the variance" [(*2*), Section 9.5].

To test whether the between-laboratory variability is significant, an F-test is performed. [(*3*), Section 8.3] A significant result provides evidence that σ_B^2 is not zero, in which case one is evidently interested in obtaining an estimate for it.

Components of Variance. Usual Estimating Procedure. The usual way of estimating σ_W^2 and σ_B^2 consists in equating the mean squares actually obtained with their expected values.

$$MS_W = \sigma_W^2$$

$$MS_B = \sigma_W^2 + \frac{\left(N - \frac{\sum n_i^2}{N}\right)}{k - 1}\sigma_B^2$$

and solving for σ_W^2 and σ_B^2.

One thus obtains the estimates

$$\hat{\sigma}_W^2 = MS_W \tag{6}$$

$$\hat{\sigma}_B^2 = \frac{MS_B - MS_W}{\left(N - \frac{\sum n_i^2}{N}\right)\Big/(k - 1)} \tag{7}$$

Estimation of Central Value μ and Its Uncertainty. Usually, the parameter μ is estimated by the quantity $\bar{y}$ calculated by the following formula (see *e.g.*, [(*2*), Section 10.5]):

$$\bar{y} = \frac{\sum_i n_i \bar{y}_i}{\sum_i n_i} = \frac{\sum_i n_i \bar{y}_i}{N} \tag{8}$$

It is also customary to derive the uncertainty of this estimate as follows. From Equation 1 we obtain

$$\bar{y}_i = \mu + L_i + \frac{\sum_j \epsilon_{ij}}{n_i}$$

(1) O. L. Davies, "Statistical Methods in Research and Production," Oliver and Boyd Publishers, London, England, 1947.
(2) K. A. Brownlee, "Statistical Theory and Methodology in Science and Engineering," J. Wiley and Sons, New York, N. Y., 1965.
(3) J. Mandel, "The Statistical Analysis of Experimental Data," Interscience, New York, N. Y., 1964.

Downloaded from pubs.acs.org/anchami/article-pdf/42/11/1194/10250950/ac60293a019.pdf by UNIV SCIENCE AND TECHNOLOGY CHINA user on 14 October 2026

Table I. Analysis of Variance

Source	DF	SS	MS	E(MS)
Within laboratories	$N - k$	$SS_W = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2$	$MS_W = \frac{SS_W}{N - k}$	σ_W^2
Between laboratories	$k - 1$	$SS_B = \sum_i n_i(\bar{y}_i - \bar{y})^2$	$MS_B = \frac{SS_B}{k - 1}$	$\sigma_W^2 + K\sigma_B^2$

where

$$\bar{y}_i = \text{average of results for lab } i = \frac{\sum_j y_{ij}}{n_i} \quad (2)$$

$$\bar{y} = \text{weighted overall average} = \frac{\sum_i n_i \bar{y}_i}{\Sigma n_i} \quad (3)$$

$$N = \text{total number of measurements} = \sum_i n_i \quad (4)$$

$$K = \left(N - \frac{\Sigma n_i^2}{N}\right) \Big/ (k - 1) \quad (5)$$

hence:

$$\text{Var}(\bar{y}_i) = \sigma_B^2 + \frac{\sigma_W^2}{n_i}$$

Using this relation, Equation 8 leads to

$$\text{Var}(\bar{y}) = \frac{1}{N^2} \sum_i n_i^2 \text{ Var}(\bar{y}_i) = \frac{\sum_i n_i^2}{N^2} \sigma_B^2 + \frac{1}{N} \sigma_W^2. \quad (9)$$

An estimate of the variance of $\bar{y}$ is therefore obtained by substituting in Equation 9, the estimates given by Equations 6 and 7.

Critique of the Usual Formulas. It is evident, from the above formula for $\bar{y}$ (Equation 8), that this estimate gives the same weight to each individual observation, so that a laboratory that makes, say, ten measurements, carries a weight five times larger than a laboratory that would have made only two measurements. This is a correct procedure, provided that there are no laboratory biases, *i.e.*, provided that $\sigma_B^2 = 0$. If, on the other hand, $\sigma_B^2 > 0$, *i.e.*, if each laboratory has its own bias L_i, the above estimate for μ weights each L_i by n_i, thus giving greater weight to the bias L_i corresponding to a laboratory that happened to have made a larger number of measurements.

The same criticism applies to the formula used to calculate SS_B, and consequently to σ_B^2. Here, the deviation $\bar{y}_i - \bar{y}$, of a laboratory average from the weighted grand average, is weighted by n_i. This weighting is correct in regard to the component in $(\bar{y}_i - \bar{y})$ due to within laboratory replication error. It is not correct in regard to the component in $(\bar{y}_i - \bar{y})$ due to the laboratory biases L_i.

It should also be observed that Equations 8 and 9 are, logically, mutually inconsistent, unless σ_B^2 is taken to be zero in Equation 9. Indeed, the weighting by n_i in Equation 8 can only be justified if we assume $\sigma_B^2 = 0$, but then Equation 9 reduces to $V(\bar{y}) = \sigma_W^2/N$.

The problem of proper weighting in situations of this type has been discussed in detail by Cochran (*4*) and by Tukey (*5*). In this paper we deal with the problem in a mathematically simpler way, using an iterative computation procedure to obtain logically consistent estimates of the parameters of interest.

(4) W. G. Cochran, *Biometrics*, **10** (1), 101–29 (1954).
(5) J. W. Tukey, *Ann. Math. Statist.*, **28**, 43–56 (1957).

ITERATIVE PROCEDURE FOR ESTIMATING COMPONENTS OF VARIANCE

In the following we will use the symbol $V(y)$ for the variance of y.

Let us represent the ratio σ_B^2/σ_W^2 by λ.

$$\lambda = \sigma_B^2/\sigma_W^2 \quad (10)$$

From the relation

$$y_{ij} = \mu + L_i + \epsilon_{ij}$$

we infer

$$\bar{y}_i = \mu + L_i + \sum_j \epsilon_{ij}/n_i \quad (11)$$

Consequently,

$$V(\bar{y}_i) = V(L) + V(\epsilon)/n_i = \sigma_B^2 + \sigma_W^2/n_i = \lambda\sigma_W^2 + \sigma_W^2/n_i$$

or

$$V(\bar{y}_i) = (\lambda + 1/n_i)\,\sigma_W^2 \quad (12)$$

Let

$$\omega_i = 1/(\lambda + 1/n_i) \quad (13)$$

Then,

$$V(\bar{y}_i) = \sigma_W^2/\omega_i \quad (14)$$

Thus, the $\bar{y}_i$ constitute a set of values whose common expected value is μ (see Equation 11) and whose relative weights are ω_i. Consequently, the best estimate for μ is: [(*3*), Section 7.2]

$$\hat{\mu} = \sum_i \omega_i \bar{y}_i \Big/ \sum_i \omega_i \quad (15)$$

Furthermore, from Equation 14 we obtain

$$V(\bar{y}_i\sqrt{\omega_i}) = \sigma_W^2$$

hence, the estimate given by the expression

$$\sum_i \omega_i(\bar{y}_i - \hat{\mu})^2/(k - 1)$$

has an expected value equal to σ_W^2.

For the latter we have a valid estimate:

$$\hat{\sigma}_W^2 = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2/(N - k) \quad (16)$$

Downloaded from pubs.acs.org/anchem/article-pdf/42/11/1194/1025095O/ac60293a019.pdf by UNIV SCIENCE AND TECHNOLOGY CHINA user on 14 October 2026

Table II. A Typical Example of Round Robin Results with Unequal Replication

Laboratory	Analytical results
1	7.87
2	7.47, 8.08
3	7.36
4	6.98, 7.42
5	7.70, 7.36, 7.41
6	7.01, 6.60, 7.03, 6.81, 6.78, 6.65, 7.02, 6.62
7	8.08, 7.96
8	7.27, 7.36, 7.31

Table III. Analysis of Variance

Source	DF	SS	MS	E (MS)
Within labs	14	0.6005	0.04289	σ_W^2
Between labs	7	3.8180	0.54543	$\sigma_W^2 + 2.5195\ \sigma_B^2$

Thus, we may write

$$\sum_i \omega_i(\bar{y}_i - \hat{\mu})^2/(k-1) = \hat{\sigma}_W^2 \tag{17}$$

where $\hat{\sigma}_W^2$ is given by Equation 16.

Our aim is to find a set of ω_i for which Equation 17 is satisfied. This can be accomplished by an iterative procedure, as follows:

Let

$$G = \sum_i[\omega_i(\bar{y}_i - \hat{\mu})^2] - (k-1)\,\hat{\sigma}_W^2 \tag{18}$$

We must find ω_i such that $G = 0$ (see Equation 17).

1. Start with a value λ_o, from which we derive the set

$$\omega_i^o = 1/(\lambda_o + 1/n_i), \text{ and } \hat{\mu}_o = \sum_i \omega_i^o \bar{y}_i / \sum_i \omega_i^o. \tag{19}$$

2. Calculate

$$G_o = \sum[\omega_i^o(\bar{y}_i - \hat{\mu}_o)^2] - (k-1)\,\hat{\sigma}_W^2 \tag{20}$$

3. Calculate $d\lambda$ such that

$$G(\lambda_o + d\lambda) = G_o + \frac{\partial G}{\partial \lambda}\,d\lambda = 0$$

Hence

$$d\lambda = -\frac{G_o}{\partial G/\partial \lambda} \tag{21}$$

4. Substitute $\lambda_o + d\lambda$ for λ_o, recalculated G, etc. The general expression for $\partial G/\partial\lambda$ can be shown to be

$$\frac{\partial G}{\partial \lambda} = -\sum_i \left[\omega_i^2\left(1 - \frac{2\omega_i}{\sum_i \omega_i}\right)(\bar{y}_i - \mu)^2\right] \tag{22}$$

As the starting value for λ, we may take the ratio $\lambda_o = \hat{\sigma}_B^2/\hat{\sigma}_W^2$, using Equations 6 and 7.

The iterative calculation process is easily programmed on a high-speed computer, and may be stopped when a further iteration would change λ by less than a predetermined amount, say by less than 0.01%.

From the final estimate of λ we obtain, using Equation 10,

$$\hat{\sigma}_B^2 = \lambda\hat{\sigma}_W^2 \tag{23}$$

Best Estimate for True Value and Its Precision. Having obtained a "best" value for λ, we obtain the corresponding weights:

$$\omega_i = 1/(\lambda + 1/n_i) \tag{24}$$

Using Equation 15 we then obtain the "best" estimate

$$\hat{\mu} = \Sigma\,\omega_i\bar{y}_i/\Sigma\omega_i \tag{25}$$

where the ω_i are given by Equation 24.

Using Equations 25 and 14, the variance of $\hat{\mu}$ is seen to be

$$V(\hat{\mu}) = \frac{1}{(\Sigma\omega_i)^2}\,\Sigma\,\omega_i^2\,V(\bar{y}_i) = \frac{1}{(\Sigma\omega_i)^2}\,\Sigma\,\omega_i^2\,\frac{\sigma_W^2}{\omega_i} = \frac{\sigma_W^2}{\sum_i \omega_i} \tag{26}$$

Thus, the estimated standard error of $\hat{\mu}$, as calculated by Equation 25, is

$$\hat{\sigma}_{\hat{\mu}} = \frac{\hat{\sigma}_W}{\sqrt{\sum_i \omega_i}} \tag{27}$$

Extreme Cases for ω_i Weighting. Two extreme weighting procedures for obtaining "best" average values have commonly been used: weighting each measurement equally, and weighting each laboratory's average equally. The above defined ω_i weighting factor [$\omega_i = 1/(\lambda + 1/n_i)$] can describe both of these limits as well as all intermediate cases. The two limits are easily demonstrated. When $\lambda = 0$, *i.e.*, when the between-laboratory component of variance is zero (see Equation 10), then $\omega_i = n_i$ and each measurement receives an equal unit weight. When λ is large with respect to each $1/n_i$, then ω_i is effectively equal to the constant $1/\lambda$. For this case, each laboratory's average receives equal weighting.

A NUMERICAL EXAMPLE

The data in Table II can be considered as a typical example of a round robin test in which the very desirable condition of equal numbers of replicates in all laboratories could not be fulfilled. In such cases, improper weighting may result in a distortion of the ratio of between to within laboratory variance, and in a poorly estimated central value.

The analysis of variance is shown in Table III.

This gives the estimates

$$\sigma_W^2 = 0.04289,\ \sigma_B^2 = 0.19946, \text{ hence } \lambda_o = \frac{0.19946}{0.04289} = 4.650.$$

Applying the iterative procedure, we obtain the new estimate $\lambda = 3.240$. This value was obtained at the ninth iteration, and differed from that of the eighth iteration by less than 0.01%.

Three possible weighting procedures suggest themselves, based on what is assumed about σ_B^2:

(a) $\sigma_B^2 = 0$, which is equivalent to $\lambda = 0$ (see Equation 10), and leads to the weighting formula $\omega_i = n_i$ (see Equation 13).

(b) σ_B^2 is estimated by Equation 7, and defines a value λ_o given by Equation 10; the weighting formula becomes $\omega_i = \dfrac{1}{\lambda_o + \dfrac{1}{n_i}}$

(c) σ_B^2 and the corresponding λ are estimated by the iterative procedure; the weighting formula becomes $\omega_i = \dfrac{1}{\lambda + \dfrac{1}{n_i}}$.

Downloaded from pubs.acs.org/anchem/article-pdf/42/11/1194/1025095O/ac60293a019.pdf by UNIV SCIENCE AND TECHNOLOGY CHINA user on 14 October 2026

Table IV. Weighting Factors for Laboratory Averages under Different Weighting Formulas

Laboratory	Absolute weights, ω_i			Relative weights, ω_i'		
	(a)[a]	(b)	(c)	(a)	(b)	(c)
1	1	0.1770	0.2358	0.0454	0.1144	0.1107
2	2	0.1942	0.2674	0.0909	0.1255	0.1255
3	1	0.1770	0.2358	0.0454	0.1144	0.1107
4	2	0.1942	0.2674	0.0909	0.1255	0.1255
5	3	0.2007	0.2798	0.1364	0.1297	0.1313
6	8	0.2094	0.2972	0.3636	0.1354	0.1395
7	2	0.1942	0.2674	0.0909	0.1255	0.1255
8	3	0.2007	0.2798	0.1364	0.1297	0.1313

[a] Weighting formulas:

(a) $\omega_i = n_i$; (b) $\omega_i = \dfrac{1}{\lambda_o + \dfrac{1}{n_i}}$; (c) $\omega_i = \dfrac{1}{\lambda + \dfrac{1}{n_i}}$

Relative weight $\omega_i' = \omega_i / \sum_i \omega_i$.

Table V. Estimates of Parameters under Different Weighting Formulas

Parameter	Weighting formula		
	$\omega_i = n_i$	$\omega_i = \dfrac{1}{\lambda_o + \dfrac{1}{n_i}}$	$\omega_i = \dfrac{1}{\lambda + \dfrac{1}{n_i}}$
λ	0	4.650	3.240
σ_W^2	0.04289	0.04289	0.04289
σ_B^2	0	0.19946	0.13898
μ	7.280	7.470	7.466
σ_μ	0.044[a] 0.204[b]	0.166	0.142

[a] Using Equation 9 with $\sigma_B^2 = 0$ (which is equivalent to $\lambda = 0$).
[b] Using Equation 9 with σ_B^2 given by Equation 7.

Table IV lists absolute and relative weights for these three weighting procedures. The relative weight is defined by $\omega_i' = \omega_i / \Sigma_i \omega_i$.

Table V lists the estimates of λ, σ_w^2, σ_B^2, μ, and σ_μ for the three weighting procedures. Several remarks are in order. The logical estimate of σ_μ corresponding to $\hat{\mu} = 7.280$ would be 0.044, not 0.204, although the latter is the commonly used one. The former, however, is the only estimate consistent with the weighting process leading to $\hat{\mu} = 7.280$. The estimate $\hat{\sigma}_{\hat{\mu}} = 0.044$ would, however, lead to a serious overestimation of the precision of $\hat{\mu}$ and to a false sense of security on the part of the experimentor. The weighting procedure $\omega_i = n_i$ is definitely not recommended, unless in the analysis of variance, $MS_B < MS_W$.

The value of λ_o used in the second weighting procedure is not the best estimate that can be obtained from the data and its shortcomings are reflected more in the estimate of σ_μ than in that of μ. The advantage of using the value λ_o is that it is readily calculated, and may well be satisfactory in many instances. If, however, the expense of obtaining the data is considerable, there is generally ample justification for using the slightly more complex iterative procedure. This is the only procedure in which the estimates of all the parameters are mutually consistent.

In cases, such as our illustration, where one of the laboratories makes an appreciably larger number of measurements than all others, the iterative procedure prevents the bias of this laboratory from distorting the end results, including the estimate of σ_B^2, and that of μ and of its standard error.

ACKNOWLEDGMENT

Thanks are due to Ernest R. Plante and William S. Horton for helpful suggestions and criticisms in conjunction with this work.

RECEIVED for review April 15, 1970. Accepted July 2, 1970.

Downloaded from pubs.acs.org/anchem/article-pdf/42/11/1194/1025095O/ac60293a019.pdf by UNIV SCIENCE AND TECHNOLOGY CHINA user on 14 October 2026